

Deal Origination Tool

Product Flow Map & Business Rules

Version 1.0 | 6 July 2026 | Prepared for print — all rules verified against production code | CONFIDENTIAL

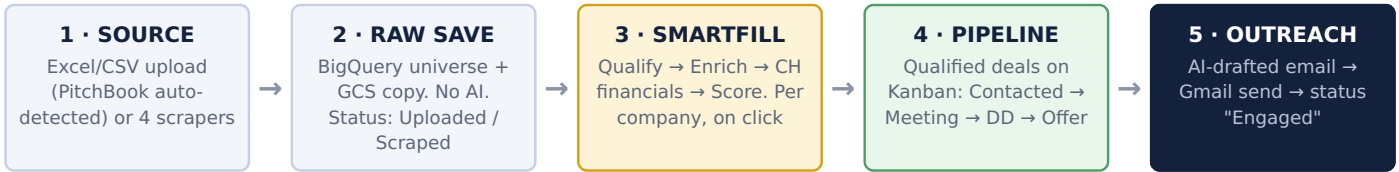
What the product does

An AI-powered platform that sources, qualifies, enriches, scores and manages B2B SaaS investment targets for Averroes Capital's private equity deal flow. Thesis: **UK/Ireland · B2B SaaS / software / tech-enabled services** · founder-led or bootstrapped preferred · target EBITDA **£1M-£10M** (revenue under £50M).

THE ONE ARCHITECTURE RULE THAT GOVERNS EVERYTHING

Ingest saves RAW data only — no AI at ingest. SmartFill is the ONLY AI layer, triggered per-company, on demand, by a button click.

The core loop



The system at a glance



Two screens

Screen	Purpose
Master Universe	Every company ever ingested. Search, filter, upload files, run SmartFill, draft outreach.
Deal Pipeline	Kanban of active deals only. Drag between stages, sorted by Fit Score.

Stack (for orientation)

Layer	Tech
Frontend / Backend	Next.js 16 · FastAPI (Python 3.13)
AI	Gemini 2.5 Flash (+ Google Search grounding)
Data / Hosting	BigQuery · GCS · Cloud Run (europe-west1)

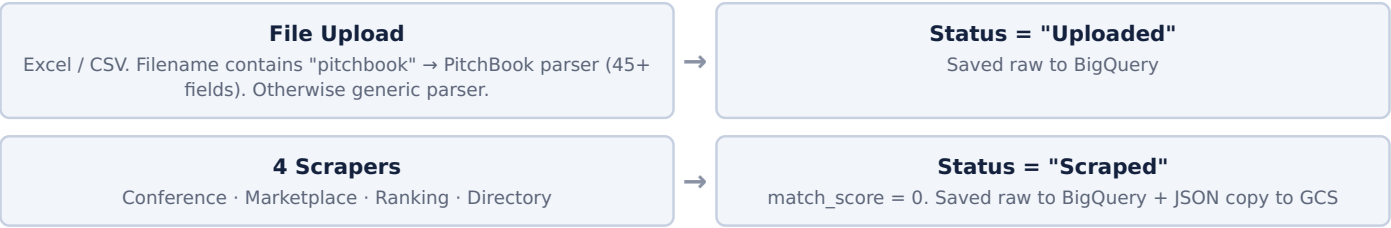
How to read this document

Pages 2–3: end-to-end flow and the SmartFill engine. Pages 4–6: the business rules in detail (qualification, Companies House, scoring). Page 7: deal lifecycle & outreach. Page 8: eight real-world scenarios showing how the rules play out. **CONDITION** tags mark points where behaviour branches.

1 · End-to-End Flow

From raw source to signed deal — what happens, in order, and where AI is (and is not) involved.

Stage A — Getting companies in (NO AI)

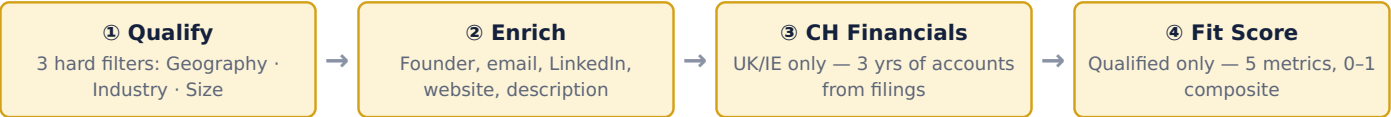


Known limitation: most scrapers currently return demo/fallback data — real sites block scraping. Uploads (PitchBook) are the reliable ingest path today.

Stage B — The universe (holding pen)

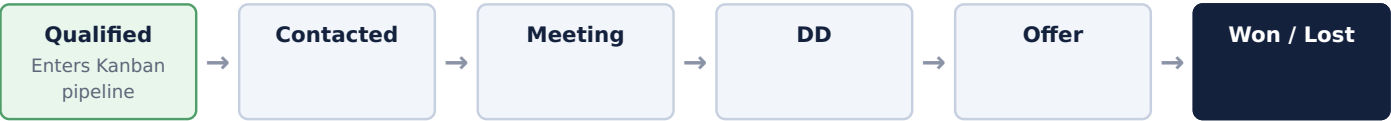
Every ingested company sits in the BigQuery **targets** table and appears on the Master Universe screen. Nothing is judged yet: no qualification, no scores, no enrichment. Companies wait here until you press **SmartFill**.

Stage C — SmartFill (the ONLY AI moment)



One click per company. Result: status flips to **Qualified** or **Not a Fit**, and every field learned along the way is written back to BigQuery.

Stage D — Working the deal



Stages move by drag-and-drop; every move and note is logged to **activity_log** with author and timestamp. AI outreach can be drafted and sent from the Universe screen at any stage.

Where each piece of data lives

Store	What it holds	Written by
BigQuery · targets	The universe: 70+ columns per company (PitchBook, CH financials, scores, contacts)	Ingest (raw) + SmartFill (everything else)
BigQuery · activity_log	Status changes and notes — the audit trail	Stage moves, notes, requalification
BigQuery · qualification_config	Versioned qualification criteria (JSON)	Criteria Chat "apply"
GCS bucket	Uploaded files, scraped JSON, CH filing PDFs (ch-filings/...)	Ingest + SmartFill step ③

2 · Inside SmartFill — The Decision Flow

What actually happens when you press the button, including every branch condition.

1

Qualification — 3 hard filters

IF Gemini key set → AI qualification, ELSE keyword-only fallback

Gemini 2.5 Flash assesses geography, industry and size in one pass, using all available data. All three must pass → **Qualified**. Any failure → **Not a Fit**. If Gemini is unavailable or errors, the system falls back to deterministic keyword matching (same filters, less inference). Full rules on page 4.

2

Enrichment — always runs

RUNS EVEN IF NOT A FIT

Gemini + Google Search discovers: founder/CEO name, contact email, LinkedIn URL, company website, and a 1-2 paragraph description. Saved regardless of qualification outcome — so even rejected companies keep useful contact data.

3

Companies House financials

ONLY IF geography filter passed (UK/IE)

Runs whenever the company is UK/Ireland — **even if it failed the industry or size filter**. Searches CH, picks the best match by scoring algorithm (rejects weak matches), downloads latest accounts PDFs, and has Gemini extract up to 3 years of Revenue / Profit / Assets / Net Assets / Cash / Employees. First PDF is stored in GCS and viewable from the company card. Full pipeline on page 5. If CH errors, SmartFill continues without it.

4

Averroes Fit Scoring

ONLY IF status = Qualified

5 metrics scored 0–1, composite = average of available metrics. Hard rule: at least **4 of 5** metrics must be assessable, otherwise the fit score is **null** ("insufficient data") — never a fake number. Full model on page 6.

5

Write-back to BigQuery — single atomic UPDATE

Status, size bucket, contacts, all CH fields, all 5 metric scores + composite, and score explanations are written in one UPDATE. **DESCRIPTION RULE** the new description only replaces the old one if it is **longer** — richer data never gets overwritten by thinner data.

Outcome matrix — what you end up with

Case	Status	Contacts	CH financials	Fit score
UK/IE + tech + right size	Qualified	✓	✓	✓ (if ≥4 metrics)
UK/IE but wrong industry or too large	Not a Fit	✓	✓ (still fetched)	—
Right industry, outside UK/IE	Not a Fit	✓	— (skipped)	—
Qualified but data too thin (<4 metrics)	Qualified	✓	✓/—	null — shows unscored

Why this design: AI cost and latency are incurred only for companies a human chose to examine. The universe can hold thousands of raw names at zero AI cost; SmartFill spends credits one company at a time.

Related batch action — "Requalify All"

Separate from SmartFill: re-runs the **keyword-based** filters across the whole universe in one batch UPDATE (fast, no per-company AI, no enrichment, no CH, no scoring). Used after changing qualification criteria to instantly re-sort the universe into Qualified / Not a Fit.

3 · Qualification — The 3 Hard Filters

All three must pass. Criteria live in BigQuery (qualification_config) and are editable in plain English via Criteria Chat; defaults shown below.

Filter 1 — Geography: UK or Ireland?

Fields checked

regionhq_countryhq_cityhq_location

Match list (any hit passes): UK, United Kingdom, England, Scotland, Wales, Northern Ireland, Ireland, London, Manchester, Birmingham, Leeds, Bristol, Edinburgh, Glasgow, Dublin, Belfast, Cambridge — plus country codes GB, UK, IE, IRL. With Gemini, HQ can also be *inferred* from context when fields are empty.

Filter 2 — Industry: technology / SaaS / software?

Fields checked

sectordescriptionkeywordsverticalsindustry_groupemerging_spaces

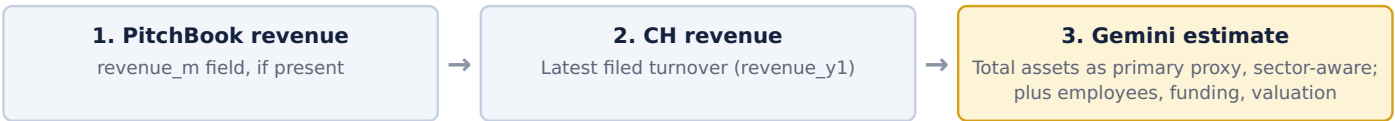
33 tech keywords (any hit passes): software, saas, platform, cloud, tech, digital, data, analytics, ai, machine learning, automation, cyber, fintech, healthtech, edtech, proptech, insurtech, regtech, legaltech, martech, adtech, devops, api, microservices, ml, deep learning, nlp, computer vision, robotics, iot, blockchain, web3, ecommerce.

Business rule: traditional industries without tech as a core product do **not** count — a logistics firm with a website is not "tech-enabled".

Filter 3 — Size: revenue under £50M?

Bucket	Revenue	Qualifies?	Typical profile
Micro	< £5M	YES	<30 employees, early stage
Small	£5M – £15M	YES	30-100 employees, bootstrapped-profitable or Series A/B
Mid	£15M – £50M	YES	100-500 employees, established product
Large	> £50M	NO	Too big for the lower-mid-market thesis

How size is determined — the cascade



Condition	Behaviour
Hard revenue number exists (steps 1-2)	Rule-based bucketing, high confidence — always preferred over AI judgment
No revenue disclosed (common: UK micro-entity accounts)	Gemini sizes from proxies. Rule of thumb baked in: tech/services revenue ≈ 2-4× total assets; capital-heavy ≈ 1-1.5×
Signals are mixed / thin	Conservative bias: lean smaller. Default bucket if undecidable: "Small" (low confidence) — i.e. benefit of the doubt, stays qualified
"Large" classification	Only with strong evidence revenue exceeds £50M — never on a hunch
Keyword-fallback mode & size unknowable	Size filter is skipped; qualification decided by geography + industry alone

Changing the rules — Criteria Chat



4 · Companies House Financials Pipeline

Runs only for UK/Ireland companies. Turns the company name into 3 years of real filed accounts.

- 1

Search the CH register
Company name queried against the official Companies House API → top 10 candidate matches.
- 2

Pick the best match — scoring algorithm **REJECT IF BEST SCORE < 15**
See scoring table below. If no candidate reaches 15 points, **no CH data is used at all** — a wrong company's accounts are worse than none.
- 3

Company profile
Official name, company number, status, SIC codes, incorporation date.
- 4

Filing history
Latest filed **accounts** located (up to 3 filings used for a 3-year picture).
- 5

Download PDFs → store in GCS
First PDF saved to GCS (ch-filings/{company number}/...) and permanently viewable via "View CH Filing PDF" on the company card.
- 6

Gemini reads the accounts
PDF sent to Gemini 2.5 Flash → extracts current + comparative year figures; de-duplicates across filings.

Match-scoring algorithm (step 2)

Signal	Points	Signal	Points
Exact name match	+100	Company status: Dissolved	−20
One name contains the other	+60	SIC/snippet tech terms align with our context	+20
Word overlap (per shared word)	+15	UK registered address	+10
Company status: Active	+30	Acceptance threshold	≥ 15

What comes out

Up to 3 years each of: **Revenue · Profit · Total Assets · Net Assets · Cash · Employees**, plus filing type, match confidence, official CH name/number/status/SIC codes, and the stored PDF path.

The micro-entity reality: most UK SMEs file micro-entity or abbreviated accounts that do **not** disclose turnover — but do disclose total assets and net assets. The system is built around this: assets become the primary sizing proxy (page 4) and the revenue-size fallback chain (page 6) keeps scoring possible.

Downstream consequence of a failed match: no CH data → no filed revenue history → the Revenue Growth metric cannot be computed → the company must hit 4/5 via the four other metrics, or its fit score is null.

5 · Averroes Fit Score — The 5-Metric Model

Qualified companies only. Each metric scores 0–1; composite = simple average of *available* metrics.

THE INTEGRITY RULES

① **Minimum 4 of 5 metrics must be assessable — otherwise fit score = null, shown as unscored.** ② **Missing metrics are excluded, never counted as zero.** ③ **AI metrics must cite evidence found via live web search — no evidence, no score.**

#	Metric	Source	How it's judged
1	Revenue Growth	Local — CH filings	YoY change, latest vs prior filed year. <–20%: 0.0 · –20–0%: 0.1–0.2 · 0–10%: 0.2–0.5 · 10–25%: 0.5–0.75 · 25–50%: 0.75–0.9 · 50%+: 0.9–1.0
2	Revenue Size	Local + AI fallback	Fit to sweet spot (see bands below); revenue found via the 7-step cascade below
3	Employee Growth	Gemini + web search	LinkedIn headcount, job postings, hiring/layoff news. Shrinking: 0–0.2 · flat: 0.2–0.4 · stable/modest: 0.4–0.6 · healthy 10–30% YoY: 0.6–0.8 · rapid 30%+: 0.8–1.0
4	Business Model Fit	Gemini + web search	B2B? SaaS/recurring revenue? Tech core? Not B2B/tech: 0–0.2 · mostly services/B2C: 0.2–0.4 · tech-enabled services: 0.4–0.6 · B2B software subscription: 0.6–0.8 · pure B2B SaaS: 0.8–1.0
5	Market Sentiment	Gemini + web search	Press, awards, reviews, brand. Negative coverage: 0–0.2 · minimal presence: 0.2–0.4 · neutral: 0.4–0.6 · positive/awards: 0.6–0.8 · strong brand & thought leadership: 0.8–1.0

Revenue Size — the 7-step fallback cascade

Try	Source	Conversion
1	Companies House filed revenue	Used directly (strongest)
2	PitchBook revenue	Used directly
3	Total assets (CH)	× 2.5 (conservative tech multiplier)
4	Estimated EBITDA	× 6.5
5	Valuation estimate	÷ 6
6	Last financing round	× 3 (rough heuristic)
7	Gemini + live web search estimate	Last resort — keeps scraped companies scoreable

Revenue Size — scoring bands (sweet spot £15–30M)

Revenue	Score	Read
< £2M	0.10 – 0.30	Too early
£2 – 5M	0.30 – 0.50	Small but watchable
£5 – 15M	0.50 – 0.75	Good
£15 – 30M	0.75 – 1.00	Sweet spot
£30 – 50M	0.75 → 0.60	Acceptable, large end (declining)
> £50M	0.30	Too big (shouldn't normally be qualified)

Reading the composite on screen

Fit score	Badge	Interpretation
≥ 0.70	GREEN	Strong thesis fit — prioritise
0.40 – 0.69	AMBER	Partial fit — review the per-metric breakdown
< 0.40	RED	Weak fit despite passing hard filters
null	UNSCORED	<4 metrics assessable — needs more data, not a judgment

Every score carries a written explanation (visible in the company drawer) citing the data or search evidence used.

6 · Deal Lifecycle & AI Outreach

The 10 statuses, who can set them, what the pipeline shows, and how outreach emails work.

All 10 statuses

Status	Set by	In Kanban pipeline?
Scraped / Uploaded	Ingest (automatic)	No — Universe only
Not a Fit	SmartFill fail · Requalify-all · manual "Remove" (also zeroes score)	No
Qualified	SmartFill pass · Requalify-all	Yes — entry column
Contacted → Meeting → DD → Offer	Manual — drag on Kanban (logged to activity_log)	Yes
Won / Lost	Manual — closes the deal	No — archived off the board
Engaged	Automatic only — set when an outreach email actually sends	Yes

Quirk to be aware of: "Engaged" is set automatically on email send and shows in the pipeline, but it is **not** in the valid list for manual stage changes — you cannot drag a card into or out of it via the standard status update, you move on by setting Contacted/ Meeting/etc. Candidate for cleanup.

Business rules around stage moves

Rule	Behaviour
Stage validation	Manual updates must be one of the 10 statuses above (excl. Engaged) — anything else is rejected
Audit trail	Every stage change and note → activity_log (old status, new status, author, timestamp)
Remove from pipeline	Sets Not a Fit and match_score = 0 — the company stays in the Universe, it is never deleted
Kanban ordering	Cards sorted by Fit Score descending — best prospects float to the top of each column

AI Outreach — draft, then send



Condition	What happens
Gemini unavailable at draft time	Falls back to a standard template — drafting never hard-fails
No recipient email on file	Send is blocked with an error — enrich first
SMTP password not configured (current state)	Every send fails with a clear error; drafting still works. Fix: set a Gmail App Password as the OUTREACH_SMTP_PASSWORD env var
Send succeeds	Status flips to Engaged (unless Not a Fit); result returned to screen

Open gaps (as of this document)

No authentication on the dashboard (public URL) · scrapers largely demo data · no CRM sync · Gmail App Password pending · deep-dive /analyze endpoint returns mock data · no bulk "SmartFill All" yet.

7 · Scenario Playouts — How the Rules Combine

Eight realistic cases traced end-to-end through the exact business rules.

S1 · PitchBook upload: London B2B SaaS, £8M revenue on file

QUALIFIED + SCORED

Upload → "Uploaded" → SmartFill: geography ✓ (London) · industry ✓ (saas) · size: hard revenue → **Small**, high confidence ✓ → **Qualified**. CH match found → 3yrs accounts + PDF stored. Revenue growth computed from filings; likely 5/5 metrics → composite score, green/amber badge. Appears in pipeline.

S2 · Scraped UK company, micro-entity accounts (no turnover disclosed)

QUALIFIED, SCORED VIA FALLBACKS

Scrape → "Scraped", score 0 → SmartFill: geo ✓ · tech ✓ · size: no revenue anywhere → Gemini sizes from **total assets proxy** (tech ≈ 2-4x assets), conservative → Small ✓ → **Qualified**. CH: match found but no turnover → Revenue Growth **unavailable**. Scoring: revenue size via assets × 2.5 (cascade step 3) + 3 web-search metrics = **4/5** → **passes the 4-metric rule** → scored.

S3 · US SaaS company in an uploaded list

NOT A FIT — GEOGRAPHY

SmartFill: geography ✗ → **Not a Fit** immediately. Enrichment still runs (contacts saved for reference). CH **skipped** (not UK/IE). Scoring skipped. Stays in Universe, never enters pipeline.

S4 · Manchester tech company, £60M revenue

NOT A FIT — TOO LARGE

SmartFill: geo ✓ · tech ✓ · size: hard revenue → **Large** ✗ → **Not a Fit**. Note: CH financials **still fetched and saved** (condition is UK/IE, not qualification) — the data is there if the thesis cap ever changes via Criteria Chat.

S5 · Generic company name — CH can't find a confident match

QUALIFIED, UNSCORED

SmartFill: qualifies on Gemini's inference, but best CH match scores < 15 points → **no CH data used**. No filed revenue → Revenue Growth unavailable. If web search also finds no employee signal: only 3/5 metrics → fit score = **null**, shows unscored. Remedy: enrich the record (correct legal name / website) and re-run SmartFill.

S6 · Criteria change: "add the Netherlands, cap at £60M"

RULES UPDATED, BATCH REQUALIFIED

Criteria Chat: Gemini edits the criteria JSON → preview shows qualified/rejected counts before committing → Apply saves a new version to BQ → optional **Requalify-All** batch re-sorts the whole universe by keywords (no AI cost). Previously-rejected Dutch companies flip to Qualified; fresh SmartFill still needed for their scores.

S7 · Outreach today (SMTP not yet configured)

DRAFT OK, SEND BLOCKED

Draft: Gemini writes a personalised email from BQ data ✓. Send: fails with a clear error — OUTREACH_SMTP_PASSWORD unset. Status unchanged. Once a Gmail App Password is configured: send succeeds → status → **Engaged** → card appears in pipeline under Engaged.

S8 · Junk company removed from the pipeline

REMOVED, NEVER DELETED

Remove button: status → **Not a Fit**, match_score → 0, action logged to activity_log. The record stays in the Universe with all its data — re-runnable through SmartFill if it ever becomes relevant again.

IF YOU REMEMBER ONLY THREE THINGS

① **No AI at ingest** — SmartFill is the only AI moment, one company per click. ② **Three hard filters decide Qualified vs Not a Fit; five metrics decide how good a fit** — and a score is null rather than guessed when <4 metrics exist. ③ **Nothing is ever deleted** — companies only change status, and every change is logged.